

	UNIVERSITY OF ADEN - FACULTY OF ENGINEERING		
	Odd SEMESTER RE- EXAMINATION, ACAD.YEAR 2024-2025		
	Student Name		Reg No
	Subject	Principle of communication systems	
	Class		
	Max Marks		
Lecturer			

NOTES: 1) ANSWER ANY FOUR QUESTIONS.
2) They can use calculator

QUESTION No 1

- Explain the generation of the PPM signal ,with necessary block diagram and waveforms representation of signal generation ?
- Mention the advantages and disadvantages of the Frequency Modulation ?
- An audio signal represented by $m(t)=10\sin(400\pi t)$ Volt , is utilized to amplitude modulate a carrier wave of : $X_c(t)=50\sin(2500\pi t)$ Volt , Assuming the modulation index is 0.8, calculate:
 - The frequencies of the sidebands.
 - The amplitude of each sideband.
 - The efficiency of the amplitude modulation.
 - A sketch of the modulated waveform and frequency spectrum.

QUESTION No 2

- What is the Quantization In details ,why we need it after sampling, explain your answer with your example?
- Draw the generation block diagram of AM (DSB-FC) with short notes about each block ?
- An analog signal is expressed by the equation:

$$m(t)=25\cos(400\pi t)+15\sin(600\pi t)+\cos(1200\pi t)$$
 Calculate:
 - The frequencies of the components in the signal.
 - The maximum frequency (f_{max}) among the components.
 - The Nyquist rate and Nyquist interval for the signal.

QUESTION No 3

- What are the advantages and disadvantages of AM-SSB (single side band) ?
- Explain the generation and recovery of the PAM (Flat Top) signal ,with necessary equations and spectrums diagrams ?
- For the given analog signal : $x(t)=3\cos(100\pi t)+10\sin(600\pi t)+\cos(800\pi t)$
 Using FDM(Frequency Division Multiplexing) with carrier frequency :
 $f_{c1}=300\text{Hz}$ - (for the $3\cos(100\pi t)$ signal)
 $f_{c2}=700\text{Hz}$ - (for the $10\sin(600\pi t)$ signal)
 $f_{c3}=1500\text{Hz}$ - (for the $\cos(800\pi t)$ signal) ,
 Determine the:
 - bandwidth of each modulating signal
 - Total bandwidth and guard band between signals
 - Draw the frequency spectrum

QUESTION № 4

- Explain the Detection of the PWM signal ,with necessary block diagram and waveforms representation of signal generation ?
- Compare between AM and FM ?
- An FM wave is expressed as:

$$x(t)=30\cos(2\pi \times 10^6 t + 5\sin(2\pi \times 2000t))$$

Determine : The Frequency Deviation , the Bandwidth and the Power of the FM wave ?

QUESTION № 5

- Draw the block diagram of basic elements of PCM?
- What are the main different between AM-DSB.FC , AM-SSB .In power transmission ,bandwidth ,waveform representation and equation representation ?
- An angle-modulated signal is represented by the equation:

$$S(t)=12\cos(2\pi \times 1.2 \times 10^6 t + 7\sin(2\pi \times 3000t)) \text{ volts}$$

Find the following:

- The power in the modulated signal.
- The frequency deviation.
- The deviation ratio.
- The transmission bandwidth.

QUESTION № 6

- What are the advantages of Modulation?
- What is multiplexing why we need it, then draw the block diagram of FDM ,TDM ?
- For the given figure complete the table (With uniform quantization):

Sampled values of Analog signal	7.7 v	13.7 v	16 v	14.3v	6.8 v
Nearest quantizing voltage					
Quantization error					
Binary code					

Then : From the binary code draw the line code using :

- polar NRZ
- Polar RZ

